MODULE 03 · FOUNDATIONS - ALL DOMAINS

Development approaches, life cycles and tailoring

Predictive, iterative, incremental and adaptive; how to choose between them; and how to fit any of them to the project in front of you.

DOMAIN Foundations EXAM WEIGHT All domains MODULE 03 of 49

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The approach is chosen, not inherited. It follows the uncertainty.

Four life cycles

LIFE CYCLE	WHAT IT DOES
Predictive	Scope, time and cost are planned early. Work flows through phases to one main delivery
Iterative	Scope is set early, but the solution is refined through repeated cycles of feedback
Incremental	Finished pieces are delivered one after another, each adding usable capability
Adaptive, or agile	Both at once: short, fixed timeboxes that deliver finished increments and refine what comes next

ON THE EXAM

Agile is iterative AND incremental. That combination is what makes it adaptive.

How the four differ

	REQUIREMENTS	ACTIVITIES	DELIVERY	AIMS TO OPTIMISE
Predictive	Fixed early	Once for the project	Single	Cost and control
Iterative	Dynamic	Repeated until right	Single	Correctness of solution
Incremental	Dynamic	Once per increment	Frequent, smaller	Speed
Adaptive	Dynamic	Repeated until right	Frequent, small	Value from feedback

ON THE EXAM

Read a scenario for two clues: how fixed the requirements are, and how often something is delivered.

Iterative or incremental?

ITERATIVE

Sketch, then refine the whole

- A rough version of everything
- Improved in each pass
- Nothing usable until the end
- Feedback shapes the solution

INCREMENTAL

Finish one piece at a time

- One complete section first
- Then the next section
- Each piece usable on arrival
- Speed to first value

THE TIEBREAKER

Iterative improves the same thing repeatedly. Incremental adds new finished things.

Uncertainty points to the approach

REQUIREMENTS ARE...	HOW TO BUILD IT IS...	USUALLY SUITS
Clear and stable	Well understood	Predictive
Clear and stable	Uncertain	Iterative, prototypes, spikes
Uncertain or changing	Well understood	Incremental or adaptive
Uncertain or changing	Uncertain	Adaptive, with short feedback loops

ON THE EXAM

The more uncertainty on either axis, the shorter the feedback loop you want.

What else shapes the choice

FACTOR GROUP	EXAMPLES
The product	Innovation, requirement certainty, ease of change, deliverable in pieces?
Safety and rules	Safety needs, regulation, approvals that need a fixed design
The project	Stakeholder availability, schedule constraints, how funding is released
The organisation	Culture, structure, agile experience, team size and location
Contracts	Fixed price favours fixed scope; outcome-based allows change

ON THE EXAM

A product that cannot be delivered in pieces, like a bridge, limits how adaptive the build can be.

When predictive is the better choice

- Requirements are **well understood and unlikely to change**.
- Change after build is **expensive or impossible**: construction, hardware, physical installation.
- Regulators or contracts need an **approved design and baseline** before work starts.
- Not because 'that is how we always do it', and not because agile feels unfamiliar.

> ON THE EXAM
>
> Predictive is not old-fashioned. It is the right tool when the answer is already known.

When adaptive is the better choice

- Requirements are **unclear, evolving or discovered** by using the product.
- The product can be **delivered in usable slices**, and early value matters.
- Customers or users can **give frequent feedback**, and the team can act on it.
- Not when nobody from the business can take part. Agile without feedback is just unplanned work.

> ON THE EXAM
>
> Adaptive needs an engaged customer. Without feedback, short cycles have nothing to learn from.

Hybrid is not one thing. It is a continuum, chosen piece by piece.

Common ways approaches are combined

PATTERN	EXAMPLE
Adaptive build, predictive roll-out	An app is built in sprints, then deployed to 40 branches on a fixed plan
Side by side	Hardware installed predictively while an agile team builds the software
Mostly predictive, adaptive parts	A building project uses iterations only for its interior design choices
Mostly adaptive, predictive parts	An agile product team with one fixed regulatory submission milestone

ON THE EXAM

Choose the approach per workstream, by its uncertainty. Then plan how the pieces integrate.

Project Lantern on the continuum

- **Predictive:** kiosk hardware, network upgrades in six clinics, data migration, the regulator's approval.
- **Adaptive:** the patient booking app, built in two-week iterations with clinic staff and patient panels.
- **Integration points:** a fixed go-live window per clinic, and a release plan that lines up with it.
- The steering committee sees one integrated plan, with a roadmap for the app and a baseline for the rest.

> ON THE EXAM
>
> One integrated plan, two ways of working inside it. That is what 'hybrid' means on the exam.

Delivery cadence

CADENCE	WHAT IT MEANS	TYPICAL FIT
Single delivery	Everything arrives at the end	Predictive construction, events
Multiple deliveries	Several planned releases or phases	Phased roll-outs, incremental builds
Periodic delivery	Releases on a fixed rhythm	A release at the end of every iteration
Continuous delivery	Increments released as soon as they are ready	Digital products with automated pipelines

ON THE EXAM

Cadence is a separate choice from life cycle. Match it to how soon users can benefit.

The agile mindset: four values

WE VALUE...	...OVER
Individuals and interactions	Processes and tools
Working software	Comprehensive documentation
Customer collaboration	Contract negotiation
Responding to change	Following a plan

> ON THE EXAM
>
> The items on the right still have value. The items on the left have more. Not 'instead of'.

Agile principles the exam leans on

- **Early and continuous delivery** of valuable work satisfies the customer.
- **Welcome changing requirements**, even late, when they give a competitive advantage.
- **Motivated, self-organising teams**, given support and trusted to get the job done.
- **A sustainable pace**, and regular reflection to become more effective.

> ON THE EXAM
>
> An answer that burns the team out or imposes their method contradicts the principles, whatever it promises.

Tailoring, in four moves

01

Select the approach

From uncertainty and the suitability factors.

02

Tailor for the organisation

Respect governance, policy and required controls.

03

Tailor for the project

Size, risk, team, stakeholders, product.

04

Keep improving

Review what works; adjust in retrospectives.

ON THE EXAM

Tailoring never ends at kickoff. The team keeps adjusting how it works as it learns.

What can be tailored

ELEMENT	FOR EXAMPLE
Life cycle and cadence	Phase gates, iteration length, release rhythm
Processes	How much change control, how formal the risk process is
Engagement	Who attends which reviews; how decisions are made
Tools	Boards, scheduling software, collaboration platforms
Methods and artefacts	Which documents, metrics and reports are produced

ON THE EXAM

Tailor with the people affected, and within organisational rules. Unagreed tailoring is just rule-breaking.

Updated agile vocabulary

- **Backlog refinement** replaces 'backlog grooming' in the Agile Practice Guide, Second Edition.
- **Daily coordination meeting** is the guide's framework-neutral name for the daily stand-up.
- Iterations are not always called sprints, and facilitators are not always called scrum masters.
- Questions may use either name. Recognise the practice, not the label.

> ON THE EXAM
>
> The exam is framework-neutral. Expect 'iteration', 'team facilitator' and 'coordination meeting' as often as Scrum terms.

Reading the approach from the question

WORDS IN THE SCENARIO	POINTS TOWARDS
Baseline, phase gate, WBS, change control board	Predictive
Iteration, backlog, product owner, increment, demo	Adaptive
Fixed regulatory milestone alongside iterations	Hybrid
No clue at all	Judge from uncertainty and the principles

ON THE EXAM

Most questions do not name the approach. Decide it from the clues before reading the options.

> A retailer wants a mobile app for a new subscription service it has never offered before. Nobody is sure which features customers will value.
>
> The marketing director wants to see something working within a month, and customers from a pilot group have agreed to try early versions.

Which approach should the project manager recommend?

A Predictive, with a thorough requirements phase to remove the uncertainty first.

B Adaptive, with short iterations and regular reviews with the pilot group.

C Incremental, with every increment's scope fixed at the start.

D Continuous flow with no reviews, so the team can release as fast as possible.

◆ PAUSE AND DECIDE

Which approach should the project manager recommend?

A Predictive, long requirements phase.

Customers do not know yet. Their needs are discovered by using the product, not by asking.

B Adaptive, short iterations, pilot reviews.

High uncertainty, early value wanted, and willing users to give feedback: all three fit.

C Incremental, fixed scope per increment.

Delivers early, but freezes scope just when feedback should be reshaping it.

D Continuous flow, no reviews.

Fast, but throws away the feedback that resolves the uncertainty.

THE SIGNAL

Unclear requirements, early value wanted, feedback available. Every clue points to adaptive.

A city council is replacing street lighting on a bridge with smart lamps, and launching a public dashboard showing energy use and air quality from the new sensors.

The lamp installation needs a road-closure permit for fixed dates. The dashboard's features are undecided, and residents' groups want a say.

How should the project manager structure the delivery approach?

A Fully predictive, so that the permit dates are protected.

B Fully adaptive, so that residents can shape the whole project.

C Predictive for everything, adding dashboard features later through change requests.

D Predictive for the installation and adaptive for the dashboard, within one integrated plan.

◆ PAUSE AND DECIDE

How should the project manager structure the delivery approach?

A Fully predictive.

Protects the permit but forces guesses about dashboard features nobody has decided.

B Fully adaptive.

Road closures on fixed dates cannot be iterated. Physical work needs a firm plan.

C Predictive, changes later.

Plans to be wrong, then pays for it through a stream of change requests.

D Each workstream by its uncertainty, one plan.

Fixed physical work planned firmly; uncertain digital work learned in cycles.

THE SIGNAL

Two workstreams, two levels of uncertainty. Choose per workstream, then integrate.

> A bank adopted Scrum for a project to meet a new reporting regulation. The requirements are set out in the regulation, the deadline is fixed, and no requirement has changed in six months.
>
> The sponsor complains that the ceremonies cost time and have produced no useful feedback. The team agrees.

What should the project manager have done?

A Assessed the project's characteristics with stakeholders when selecting the approach.

B Trained the team more thoroughly in Scrum before starting.

C Used shorter sprints to create more opportunities for feedback.

D Asked the PMO to mandate Scrum so the sponsor could not object.

◆ PAUSE AND DECIDE

What should the project manager have done?

A Assessed the project's characteristics when selecting.

Fixed, known requirements point to predictive. A suitability assessment would have shown it.

B Trained the team better.

The team is not the problem. The approach does not fit the work.

C Shorter sprints.

More feedback loops on requirements that never change adds cost, not value.

D Have the PMO mandate Scrum.

Uses authority to silence a valid concern. Approach follows the work, not a mandate.

THE SIGNAL

Agile is not automatically better. Requirements fixed by law, with no feedback to gain, suit a predictive plan.

> An insurer's standard method requires a formal design review board before any release. The board meets monthly.
>
> A product team now releases small increments every two weeks. Each release waits up to three weeks for the board, and customers are noticing the delays.

What should the project manager do?

A. Release increments without the board's review, since the risk of each one is small.

B. Keep waiting for the monthly board, because it is organisational policy.

C. Work with the board to agree a tailored review that meets its control objectives at the team's cadence.

D. Ask the sponsor to have the design review requirement removed.

◆ PAUSE AND DECIDE

What should the project manager do?

A Skip the review.

Breaks a governance rule unilaterally. Small risk is the board's call, not yours.

B Keep waiting.

Complies, but accepts lost value when the rule could be adapted properly.

C Agree a tailored review with the board.

Tailors for the organisation: keeps the control's purpose, fits the cadence.

D Ask the sponsor to remove it.

Goes around the board, and removing a control is not the same as fitting it.

THE SIGNAL

When a control fights the cadence, redesign the control with its owners. Do not skip it or suffer it.

> A museum is designing a new permanent gallery that will open once, on a fixed date. Curators are unsure how visitors will respond to the layout and displays.
>
> The team builds rough mock-ups, tests them with visitor groups, and reworks the whole design several times before construction begins.

Which life cycle best describes the design work?

A Iterative

B Incremental

C Predictive

D Continuous delivery

◆ PAUSE AND DECIDE

Which life cycle best describes the design work?

A Iterative

The whole design is refined through repeated feedback, with one final delivery.

B Incremental

Nothing finished is delivered in pieces. Visitors see nothing until opening day.

C Predictive

The design is deliberately reworked, not fixed early and followed.

D Continuous delivery

There is one opening, not a stream of releases.

THE SIGNAL

Repeated refinement of the whole, one delivery at the end: iterative, not incremental.

A predictive project is rolling out a customer relationship system. The core platform is on track, but the sales team's reporting needs keep changing as they use the first release.

Twenty change requests about reports are waiting for the change control board, which meets every three weeks.

What should the project manager do next?

A Continue processing each report change through the change control board.

B Freeze the reporting requirements until the platform roll-out is finished.

C Switch the whole project to Scrum immediately.

D Propose to stakeholders an adaptive approach for the reporting workstream, within existing governance.

◆ PAUSE AND DECIDE

What should the project manager do next?

A. Keep using the board for each change.
The process works, but a three-week loop cannot keep pace with needs that change weekly.

B. Freeze reporting requirements.
Stops the churn by stopping the learning. Users get reports they no longer want.

C. Switch everything to Scrum now.
The platform is on track. Changing its approach adds risk for no gain.

D. Adaptive for reporting, within governance.
Moves only the uncertain workstream along the continuum, with agreement.

THE SIGNAL

The evidence changed for one workstream only. Move that workstream, not the whole project.

Choose by uncertainty, per workstream, then tailor with the people affected.